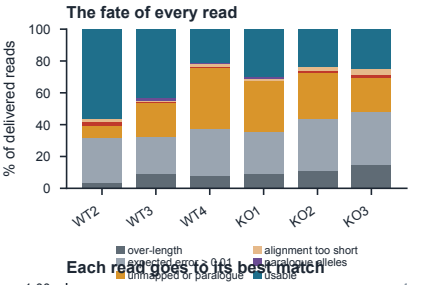
How SHMtrace turns reads into calls

Every rule the engine applies, and the same rule working on the J_H4 libraries (3 wild-type and 3 *Aicda*$^{-/-}$ mice, 500 reads each).

1 Read preparation

Every read is gated before it is used: too long for the amplicon, or noisier on average than one error in a hundred bases, and it is dropped.

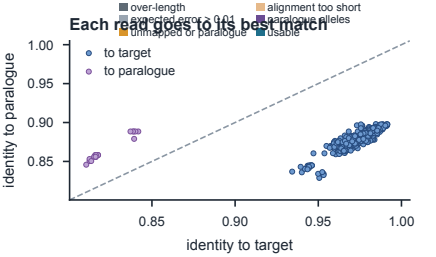
$$\bar{E} = \frac{1}{L}\sum_{i=1}^{L} 10^{-Q_i/10} \;\le\; 0.01$$



2 Competitive alignment

Each read is aligned against the target, its paralogue and any contaminant at once, and goes to whichever it matches best. A paralogue can be a target of its own (run_targets).

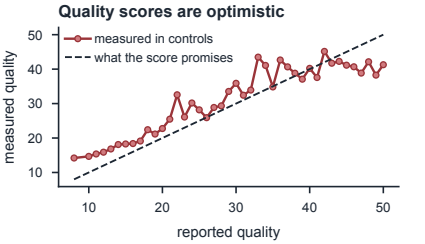
$$t^{*}(r) = \arg\max_{t}\ \text{score}(r,\ t) \quad \text{over the target, its paralogue and every contaminant}$$



3 Background measured in the controls

The error rate is not assumed from the quality scores: it is measured in the AID-null libraries, per position, substitution, strand and quality, then inflated by one factor fitted by leaving each control out in turn.

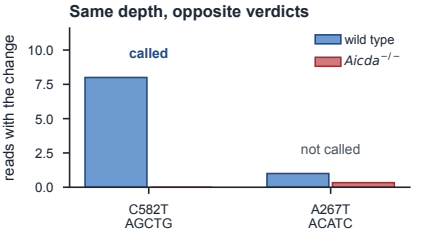
$$\hat{q} = \kappa\,\frac{\#\{\text{miscalls } x\to a,\ \text{strand } s,\ Q\ge q\}}{\#\{\text{control bases at } x,\ s,\ Q\ge q\}}$$



4 One test per candidate

Supporting reads are ranked by quality and every depth of evidence is tested against that measured rate; the best test is kept and corrected for having tried them all. A change carried by half the reads is germline, not a mutation.

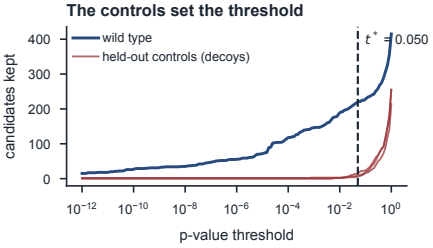
$$\lambda_j = \sum_{s} n_s\,\hat{q}_s(\ge q_{(j)}),\quad p = k\min_{j\le k} P(X\ge j),\ X\sim\text{Poisson}(\lambda_j)$$



5 Threshold set by the controls themselves

Each control is called against the other controls, giving decoy p-values. The p-value threshold is the largest at which the false calls the decoys predict are at most 5% of the calls kept.

$$\widehat{FP}(t) = \frac{1}{|D|}\sum_{d\in D}\big|\{p^{d}\le t\}\big|\max\!\left(1,\tfrac{B}{B_d}\right),\quad t^{*} = \max\{t : \widehat{FP}(t)\le\alpha\,|\{p\le t\}|\}$$



6 One number per animal

Calls are summarised per library and compared between groups on the log scale, with one value per animal, so clones of one lineage cannot stand in for independent mice.

$$\rho = \frac{\#\,\text{unique calls}}{\#\,\text{analysed bases}}\times 10^{5},\quad \text{Welch } t \text{ on } \log_{10}\rho,\ n = \text{animals}$$

